MAE 1001:
Bias in Dice: Your first experiment

Introduction to Mechanical and Aerospace Engineering

Course website: https://gwu-mae1001.github.io/fall2026/

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Photo: Kartik Bulusu



"Study hard what interests you the most in the most undisciplined, irreverent and original manner possible."

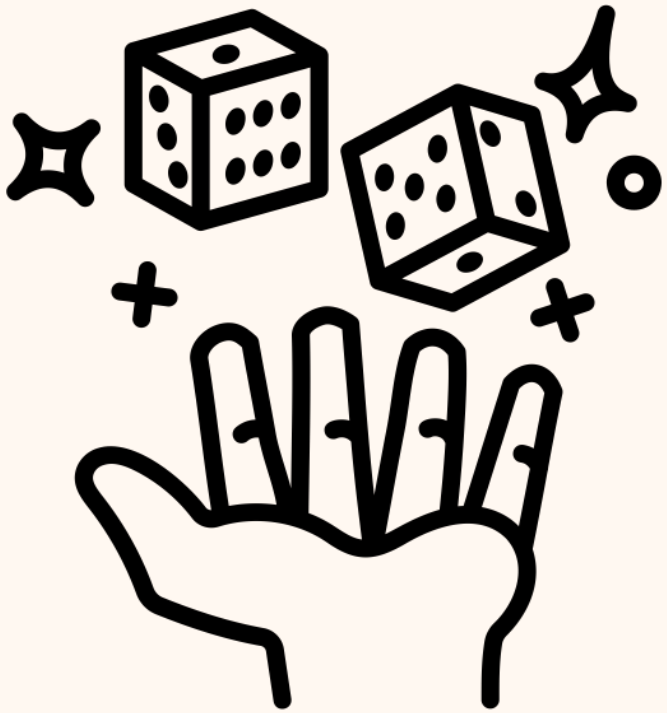
— Richard Feynmann

What is Dice Bias?

Source:
magic dice by Fran Couto from https://thenounproject.com/browse/icons/term/magic-dice Noun Project (CC BY 3.0)

Dice bias:

Happens when certain outcomes occur more often than chance would predict



Causes:

- Manufacturing flaws
- Uneven density
- Off-center markings
- Wear and tear

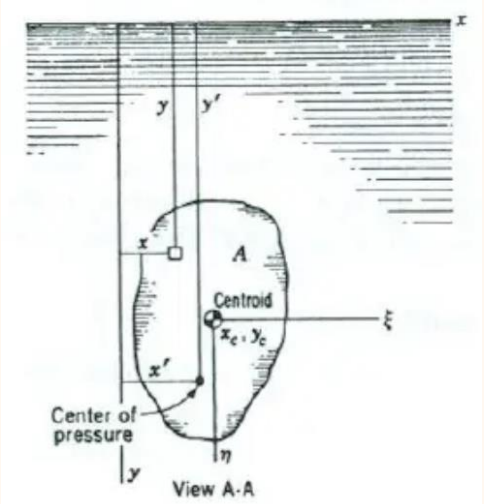
Preliminaries

Source:
https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/center-of-gravity/
https://en.wikipedia.org/wiki/Center_of_pressure_(fluid_mechanics)
Shames, I., *Mechanics of Fluids*, McGraw-Hill, New York, U.S.A
https://www.grc.nasa.gov/www/k-12/WindTunnel/Activities/buoy_Archimedes.html
https://www.britannica.com/science/buoyancy
https://en.wikipedia.org/wiki/Buoyancy
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Center of mass

The **center of mass** is a hypothetical point where the entire mass of an object may be assumed to be concentrated to visualize its motion.

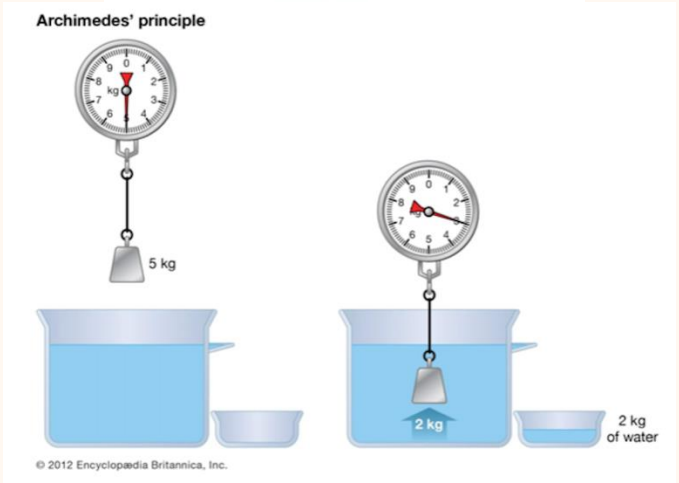
- Describes the motion of any object through space in terms of the **translation** and/or **rotation**
- **Application:** If the object is confined to rotate about some other point, like a hinge, we can still describe its motion.



Buoyancy principle:

An object floats when the upward buoyant force equals or exceeds the downward force from its weight.

The **buoyancy principle**—formulated as Archimedes' principle—states that the buoyant force acting on an object immersed in a fluid is equal to the weight of the fluid displaced by that object.



Interplay between Center of mass, Buoyancy and Dice

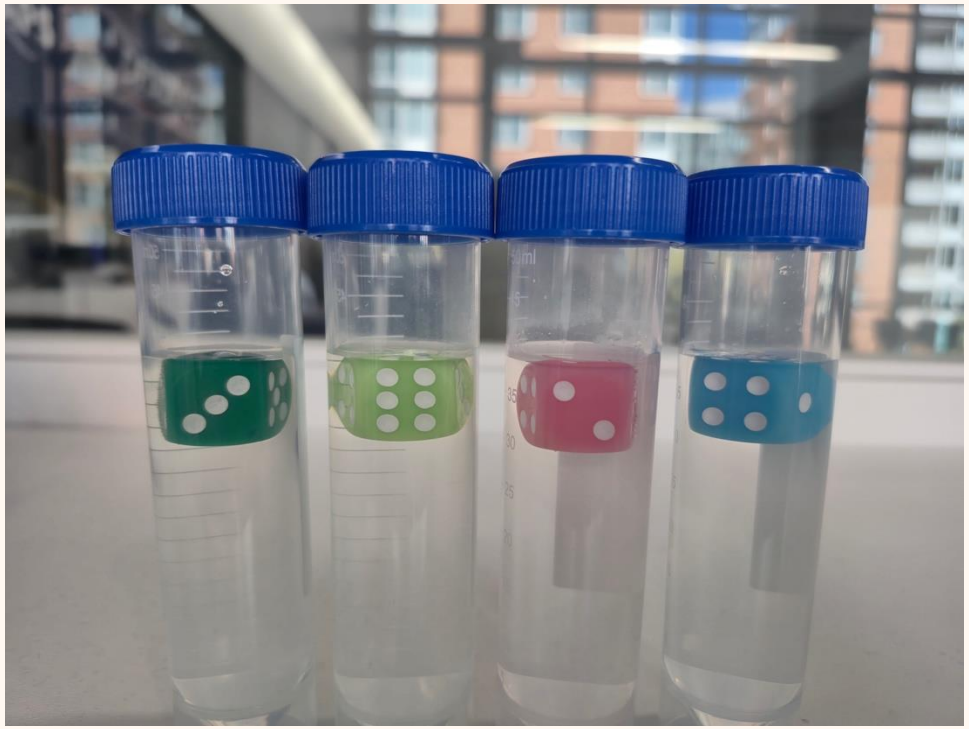
Physical principle:

- For a fair die, center of mass is at the geometrical center
- Non-uniform dice misbehave when floating due to uneven weight

Method:

- Float die in a saltwater/sugar solution to test center of mass location by observing which face points downward/upward.

A perfectly balanced die should show random orientation after several tests



Buoyancy of Float Test Method

Materials:

- Dice
- Water and salt (or sugar for higher density)
- Sealed centrifuge tube with a dice floating in it

Consistent results indicate off-center mass or bias in dice



Steps:

1. You will be given a centrifuge tube with a floating dice in it: **Shake the centrifuge tube.**
2. **Observe the dice as it floats up.**
3. **Record the face value** that shows up as the dice floats and rests on the fluid surface.
4. **Repeat the test several times for reliability.**
5. **Locate the data saved in a CSV file on your computer**

What Each Python Library Does

The imports at the top of the notebook

LIBRARY	WHAT IT DOES	OUTPUT IN THIS PROGRAM
os `import os`	Works with files on the computer (or in Colab).	**True** / **False**: does this die's CSV already exist?
datetime `from datetime import datetime`	Reads today's date.	File-name date: o_2 on Oct 23, 2026 → **o2_102326.csv**
pandas `import pandas as pd`	Holds data as a table (a **DataFrame**) and reads/writes CSV files.	**The CSV file**, one row per roll: Last Name, First Name, Die Name, Face Value
plotly.express `import plotly.express as px`	Draws interactive charts from a table.	**The histogram**: count of each face, 1–6
dash `import dash + html, dcc, Input, Output, State, ctx`	Turns Python into a web page. **html** = headings and buttons, **dcc** = text boxes and graph, **callback** + **ctx** = run code when a button is clicked.	**The dice entry form** in your browser or Colab, updating after every click

Run the Program in Google Colab

No installation — it runs in your browser

1 Open Google Colab

Go to **colab.research.google.com** and sign in with your Google account.

2 Upload the notebook

File → **Upload notebook** → Browse, and pick **BiasInDice_PythonProgram_2026.ipynb**

3 Install Dash (once)

Click **+ Code** to add a cell at the top, type the line below, press **Shift+Enter**.

```
!pip install dash
```

4 Run the program

Click ▶ on the program cell (or **Runtime** → **Run all**).
The dice entry form appears below the cell.

5 Record your rolls

Type your last name, first name and die name (e.g., **o_2**), then click the face value **1–6** after each float test.

6 Download your CSV

Open the **Files** panel (folder icon, left), find **o2_102326.csv**, then ⋮ → **Download**.

Heads up: Colab deletes your files when the session ends — download the CSV before you close the tab, then upload it to Blackboard.

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Discussion



- **How many of your felt that your dice was indeed biased?**
- **Did we do enough tests to confirm or bias or no-bias?**
- **What are the limitations of this experiment?**
- **How can we link the physical phenomena and the deeper analysis?**

- **Bouyancy reveals** a physical imbalance, but actual roll data is needed for quantification.
- **We need** a large dataset to link the physical phenomenon to mathematical analysis.
- **Assignment deliverable:** Upload your data file (CSV file) on BlackBoard to get a grade

We will repeat this test and apply statistical analysis using the chi-square test

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

Where O_i is observed frequency, E_i is expected (n faces)

Reflect & Report

5 minutes · on your own

0–1 min Look at your histogram → **1–4 min** Write answers to the four prompts → **4–5 min** Report your verdict

1 Observe

Which face showed up most often? How many tests did you record?

2 Judge

Is your die biased? What in your data makes you think so?

3 Explain

How could center of mass or uneven density cause what you saw?

4 Improve

What would make your result more trustworthy next time?

Your report

My die _____ is / is not biased.

Evidence: face __ came up __ of __ times.

Likely cause: _____

Next time I would: _____

Say your verdict to your table, then submit it with your CSV on Blackboard.